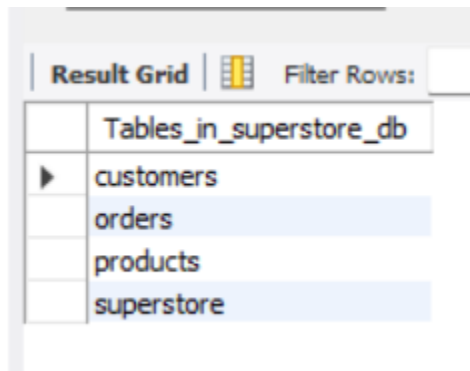


Week 3 TASK: Use Subqueries, CTEs, and Window Functions to analyze sales data from the Superstore dataset

Step 1: Setup Data

Import the Superstore dataset into a table named **superstore_raw**.

2. Create these 3 tables from it:
 - a. customers
 - b. orders
 - c. products
3. Insert data into these tables using **SELECT DISTINCT**.



Step 2: Perform Required Queries

Write and execute SQL queries for each of the following:

1. Find all orders where sales are greater than the average sales. (*Subquery*)

Result Grid Filter Rows: Exports: Wrap Cell Content: Fetch rows:													
	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region
▶	1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South
	2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South
	4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South
	8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West
	11	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West
	12	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West
	14	CA-2016-161389	12/5/2016	12/10/2016	Standard Class	IM-15070	Irene Maddox	Consumer	United States	Seattle	Washington	98103	West
	17	CA-2014-105893	11/11/2014	11/18/2014	Standard Class	PK-19075	Pete Kriz	Consumer	United States	Madison	Wisconsin	53711	Central
	25	CA-2015-106320	9/25/2015	9/30/2015	Standard Class	EB-13870	Emily Burns	Consumer	United States	Orem	Utah	84057	West
	28	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East
	36	CA-2016-117590	12/8/2016	12/10/2016	First Class	GH-14485	Gene Hale	Corporate	United States	Richardson	Texas	75080	Central
	39	CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SN-20710	Steve Nguyen	Home Of...	United States	Houston	Texas	77041	Central

2. Find the highest sales order for each customer. (Subquery)

Result Grid		Filter Rows:		Export:		Wrap Cell Content:							
	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region
	2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South
	4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South
	11	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West
	25	CA-2015-106320	9/25/2015	9/30/2015	Standard Class	EB-13870	Emily Burns	Consumer	United States	Orem	Utah	84057	West
	28	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East
	36	CA-2016-117590	12/8/2016	12/10/2016	First Class	GH-14485	Gene Hale	Corporate	United States	Richardson	Texas	75080	Central
	39	CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SN-20710	Steve Nguyen	Home Office	United States	Houston	Texas	77041	Central
	55	CA-2016-105816	12/11/2016	12/17/2016	Standard Class	JM-15265	Janet Molinari	Corporate	United States	New York City	New York	10024	East
	58	CA-2016-111682	6/17/2016	6/18/2016	First Class	TB-21055	Ted Butterfield	Consumer	United States	Troy	New York	12180	East
	68	CA-2014-106376	12/5/2014	12/10/2014	Standard Class	BS-11590	Brendan Sweed	Corporate	United States	Gilbert	Arizona	85234	West
	110	CA-2015-129476	10/15/2015	10/20/2015	Standard Class	PA-19060	Pete Armstrong	Home Office	United States	Orland Park	Illinois	60462	Central
	146	CA-2015-110744	9/7/2015	9/12/2015	Standard Class	HA-14920	Helen Andreaa	Consumer	United States	Pasadena	California	91104	West

3. Calculate total sales for each customer. (CTE)

Result Grid	Filter Rows:	Export:
Customer ID	Customer Name	Total_Sales
CG-12520	Claire Gute	1148.7800000000002
DV-13045	Darrin Van Huff	1119.4830000000002
SO-20335	Sean O'Donnell	2602.5755000000001
BH-11710	Brosina Hoffman	6255.3510000000001
AA-10480	Andrew Allen	1782.872
IM-15070	Irene Maddox	4930.4739999999999
HP-14815	Harold Pawlan	1990.314
PK-19075	Pete Kriz	8158.654
AG-10270	Alejandro Grove	2565.25799999999994
ZD-21925	Zuschuss Donatelli	1493.944
KB-16585	Ken Black	2742.75800000000003
SF-20065	Sandra Flanagan	763.54599999999998
EB-13870	Emily Burns	2767.219

4. Find customers whose total sales are above average. (CTE + Subquery)

Result Grid			
		Filter Rows:	Export:
	Customer ID	Customer Name	Total_Sales
▶	BH-11710	Brosina Hoffman	6255.351000000001
	IM-15070	Irene Maddox	4930.473999999999
	PK-19075	Pete Kriz	8158.654
	TB-21520	Tracy Blumstein	4730.628
	MA-17560	Matt Abelman	4280.620999999999
	SN-20710	Steve Nguyen	3127.9592000000007
	LC-16930	Linda Cazamias	4481.162
	RA-19885	Ruben Ausman	3832.3139999999994
	ES-14080	Erin Smith	4648.516
	KM-16720	Kunst Miller	4909.4720000000001
	KD-16270	Karen Daniels	8282.358
	JE-15745	Joel Eaton	6400.454999999999
	SC-20770	Stewart Carmic...	4492.656999999999

5. Rank all customers based on total sales. (Window Function)

Result Grid				
		Filter Rows:	Export:	Wrap Cell Cont
	Customer ID	Customer Name	Total_Sales	Sales_Rank
▶	SM-20320	Sean Miller	25043.05	1
	TC-20980	Tamara Chand	19017.847999999998	2
	RB-19360	Raymond Buch	15117.339	3
	TA-21385	Tom Ashbrook	14595.62	4
	AB-10105	Adrian Barton	14355.610999999997	5
	SC-20095	Sanjit Chand	14142.333999999999	6
	KL-16645	Ken Lonsdale	14071.917	7
	HL-15040	Hunter Lopez	12873.297999999999	8
	SE-20110	Sanjit Engle	12209.4380000000002	9
	CC-12370	Christopher Conant	12129.072	10
	TS-21370	Todd Sumrall	11885.871	11
	GT-14710	Greg Tran	11820.119999999997	12
	BM-11140	Beckv Martin	11609.9000000000001	13

6. Assign row numbers to each order within a customer. (*Window Function + PARTITION BY*)

Result Grid



Filter Rows:

Export:



Wrap Cell Content

	Customer ID	Customer Name	Order ID	Sales	Row_Num
▶	AA-10315	Alex Avila	CA-2016-103982	3930.072	1
	AA-10315	Alex Avila	CA-2014-128055	673.568	2
	AA-10315	Alex Avila	CA-2016-103982	431.976	3
	AA-10315	Alex Avila	CA-2017-147039	362.94	4
	AA-10315	Alex Avila	CA-2014-128055	52.98	5
	AA-10315	Alex Avila	CA-2016-103982	41.72	6
	AA-10315	Alex Avila	CA-2015-121391	26.96	7
	AA-10315	Alex Avila	CA-2014-138100	14.94	8
	AA-10315	Alex Avila	CA-2014-138100	14.56	9
	AA-10315	Alex Avila	CA-2017-147039	11.54	10
	AA-10315	Alex Avila	CA-2016-103982	2.304	11
	AA-10375	Allen Arnold	CA-2016-131065	499.98	1
	AA-10375	Allen Arnold	CA-2015-140921	149.97	2

7. Display top 3 customers based on total sales. (*Window Function*)

Result Grid


Filter Rows:

Export:


Wrap Cell Content:
☐

	Customer ID	Customer Name	Total_Sales	Sales_Rank
▶	SM-20320	Sean Miller	25043.05	1
	TC-20980	Tamara Chand	19017.847999999998	2
	RB-19360	Raymond Buch	15117.339	3

Step 3: Final Combined Query

Write **one final query** that shows:

- Customer Name
- Total Sales
- Rank

(Use **JOIN + CTE + Window Function** together)

Result Grid	Filter Rows:	Export:	Wrap
	Customer Name	Total_Sales	Sales_Rank
▶	Sean Miller	25043.05	1
	Tamara Chand	19017.847999999998	2
	Raymond Buch	15117.339	3
	Tom Ashbrook	14595.62	4
	Adrian Barton	14355.610999999997	5
	Sanjit Chand	14142.333999999999	6
	Ken Lonsdale	14071.917	7
	Hunter Lopez	12873.297999999999	8
	Sanjit Engle	12209.438000000002	9
	Christopher Conant	12129.072	10
	Todd Sumrall	11885.871	11
	Greg Tran	11820.119999999997	12
	Becky Martin	11609.900000000001	13
	Seth Vernon	11420.645000000007	14